PLTW POE

Activity 2.3.1 AI Robotic Greetings

PLTW

What Is Artificial Intelligence?

GOALS	MATERIALS	RESOURCES

- Gain awareness of artificial intelligence and machine learning.
- Understand the three main types of machine learning.
- Use a supervised machine learning algorithm to train a robot to accomplish a specific task.

GOALS	MATERIALS	RESOURCES

- VEX® V5 POE Kit

GOALS	MATERIALS	RESOURCES

- **Robotic Greetings Example Code**
- **Activity 2.3.1 AI Robotic Greetings (Downloadable PDF)**

Artificial Intelligence (AI) is a rapidly growing field that involves the research and development of computer systems that simulate human intelligence in machines. These machines are programmed to think like humans and mimic their actions, including the ability to learn and make decisions. Examples include computers that can easily recognize and remember objects and faces, translate between languages, and detect fraudulent credit card behavior.

Machine learning is a subfield of AI. The term broadly describes the variety of ways that computers are given the ability to learn. Traditionally, robots and other intelligent systems are programmed to complete a series of steps. In some cases, writing a program for a machine is time-consuming or impossible. For example, programming a computer to recognize handwritten text from a variety of authors is challenging in machine learning. While humans can read different handwriting and still recognize words easily, it's difficult to tell a computer how to do it. Using machine learning techniques, computers learn to program themselves.

The three main types of machine learning are supervised, unsupervised, and reinforcement. Within each category of machine learning, different algorithms are used to achieve the desired machine learning goal. In this activity, you will use a **supervised machine learning** algorithm known as **k-nearest neighbors (KNN)** to teach your robot how to perform a task.

Term	Definition
supervised machine learning	Supervised machine learning is an algorithm that is trained with labeled data sets, which allow the models to learn and grow more accurate over time. For example, an algorithm would be trained with pictures of dogs and other things, all labeled by humans, and the machine would learn ways to identify pictures of dogs on its own. Supervised machine learning is the most common type used today.
unsupervised machine learning	Unsupervised machine learning is an algorithm that searches for patterns in unlabeled data. Unsupervised machine learning can find patterns or trends that people

Term	Definition
	aren’t explicitly looking for. For example, an unsupervised machine learning program could look through online sales data and identify different types of clients making purchases.
reinforcement learning	Reinforcement learning trains machines through trial and error to take the best action by establishing a reward system. Reinforcement learning can train models to play games or train autonomous vehicles to drive by telling the machine when it made the right decisions, which helps it learn over time what actions it should take.

Your Challenge

In this activity, you will train a robot to give you a high-five, fist bump, handshake, wave, or another physical greeting. Instead of using a conditional statement to trigger the motion of your robot, you will teach your robot when to be in a "waiting for greeting" state, and when to move to the "action" state to make physical contact or greet a person.



1. Decide what type of greeting your robot will give.
2. Download the **[Robotic Greetings Example Code]** and open it in **[VEXcode V5]**.

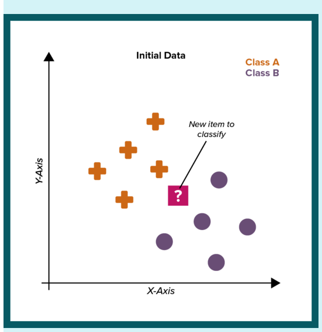
3

Review the machine-learning program your robot will use.

K-Nearest Neighbors (KNN) Algorithm

The program uses a k-nearest neighbors (KNN) algorithm to predict behavior. First, the program prompts the user to collect two sets of initial data: a “waiting for greeting” set and a “greeting” set. Each data set (also referred to as a case or a class), has five data points; each data point has a motor position and a distance sensor reading.

After the initial data is collected and classified, a new data item from the sensor is added (Figure 1). The algorithm loops to compare the new item’s sensor reading with the readings in the data sets. It tallies the number of items closest to the new item, or the “k-nearest neighbors” (Figure 2). Based on the highest number of nearest neighbors—the K value (Figure 3), the item is classified as either Class A or Class B (Figure 4). When classified, the new item sets its behavior based on the behavior of its new class.



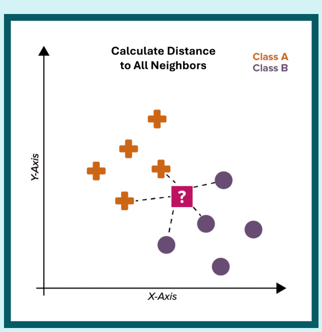


Figure 1. Initial Data with New Item

Figure 2. Distance from New Item to Neighbors

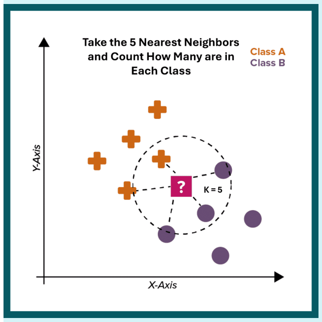


Figure 3. Nearest Neighbor Count

Figure 4. New Item Classified

4

Your robot must be able to work with the provided machine-learning program. Design it to meet the following requirements:

- Always start the robot in the waiting position. The brain interprets the starting motor position to be 0°, regardless of the motor's previous state or its physical configuration.
- Your robot arm must change from 0° in the waiting state to –75° in the greeting state. This angle is represented as the *angle_diff* variable in code.
- The distance between the waiting state and the greeting state must be at least 10 inches. This

distance is represented as the *k_dist* variable in code.

- Use a VEX® V5 Smart Motor, Distance Sensor, and Bumper Switch.
- Configure your robot's components using the ports defined in the program. An example device configuration is shown in Figure 5.

Figure 5. VEX Device Configuration

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5 Construct your greeting robot. Make sure to include the VEX V5 Distance Sensor and the Bumper Switch in your robot build.

6 Before you use the machine-learning program, use the test code to confirm that your robot is configured correctly for each position: waiting and greeting.

In **VEXcode V5**, replace the comment *# paste test code here* with the following block of test code.

```
# confirm motor positions, press button to test
brain.screen.print("Press button to callibrate motor positions")
while not bumper_a.pressing():
    wait(0.25, SECONDS)
brain.screen.clear_screen()
brain.screen.set_cursor(1,1)
```

```
brain.screen.print("Robot should be in waiting position: ")
brain.screen.set_cursor(2,1)
brain.screen.print("    ", motor_9.position(DEGREES), "degrees")
wait(2, SECONDS)
motor_9.spin_to_position(angle_diff, DEGREES)
brain.screen.set_cursor(3,1)
brain.screen.print("Robot should now be in  greeting position")
brain.screen.set_cursor(4,1)
brain.screen.print("    ", motor_9.position(DEGREES), "degrees")
wait(2, SECONDS)
brain.screen.set_cursor(5,1)
brain.screen.print("Robot should return to waiting position")
motor_9.spin_to_position(0, DEGREES)
brain.screen.set_cursor(6,1)
brain.screen.print("Test complete, stop program")
while not bumper_a.pressing():
    wait(0.25, SECONDS)
```

7. Run the test and confirm that the angle defined in *angle_diff* works with your robot design. If not, adjust your robot so its waiting angle is 0° and its greeting angle is –75°.

   ERROR TOLERANCE: Notice that when you set a position with the Smart Motor, it is not as precise as the servo motor. You may see a variation of ± 2°. This is an acceptable tolerance for your greeting robot.

8. When you have confirmed that your robot's waiting and greeting angles are correct, delete the block of test code.

9. Run the program to train your robot.

- Teach your robot five “waiting” positions. When your robot is waiting to greet someone, its arm needs to be located away from where the greeting will take place. Move your robot's arm into a waiting position.
- Enter Class A data. Press the bumper switch five times to indicate that five waiting positions have been trained.
- Teach your robot five “greeting” positions. When your robot is greeting someone, its arm needs to be located where the greeting is happening. Move your robot's arm into a greeting position. Then, hold your hand in front of the distance sensor as if you were approaching the robot with a greeting.
- Enter Class B data. Press the bumper switch five times to indicate that five greeting positions have been trained.

NOTE: Your training data will not be saved. Each time you run the code, you need to retrain your robot.

10. Test your robot. If it does not work as you wanted or expected, try retraining your robot. Pay special attention to the motor position and distance sensor values you teach your robot the waiting and greeting positions.

11. Iterate your physical robot build and your training model until you are satisfied with your robot’s function.

Real-World Robotics

Since 1837, John Deere of Deere & Company has been a pioneer in agricultural equipment. In the beginning, the company focused on plows. As time progressed, they began to produce various types of other farming equipment that enabled farmers to be more productive. Today, John Deere is focused on using artificial intelligence technology to help farmers tackle jobs on their farms.

John Deere recently debuted a fully autonomous tractor that knocks out time-consuming tillage work, which prepares land for growing crops. For the tractor to run without an operator, the tractor uses 360-degree cameras that triangulate distance and visualize objects. A high-speed processor evaluates every image that streams in from the cameras. Then, artificial intelligence sorts the images using a neural network and determines whether an area is safe to drive over. All of this happens in about 0.1 second!

The autonomous tractor can drive itself onto a field and operate entirely on its own. Such capabilities allow farmers to remain productive with limited help. While John Deere has found great success in its innovative, autonomous tractor, the company's artificial intelligence ambitions continue to grow. A robotic revolution is on the horizon, with John Deere's autonomous agricultural equipment driving the charge.[1]

John Deere Autonomous 8R Tractor video[2]

Conclusion

Question 1

How is the code you used to teach your robot different from the code you wrote in Lesson 2.2 Robots in Action?

Question 2

What are some examples of artificial intelligence that you've interacted with in your everyday life?

Question 3

How do you think AI and machine learning could help solve some of the major problems facing the world today?

Sources

Third-party images, videos, and Historical Highlight information used throughout this activity were drawn from the following sources:

1. John Deere. (n.d.). Putting Autonomy to Work. Autonomy. https://about.deere.com/en-us/our-company-and-purpose/technology-and-innovation/autonomy
2. John Deere. (2022, January 4). *Autonomous 8R Tractor | John Deere Precision AG* [Video]. YouTube. https://www.youtube.com/watch?v=QvFoRk4JsPc